

Problem Statement No. 32

AI-Powered Athlete Performance Analysis Assistant

The Challenge

Athletes generate large amounts of performance data during training sessions and competitions, including fitness metrics, match statistics, endurance levels, speed, workload indicators, and recovery information. Coaches often find it difficult to transform this raw data into meaningful insights that help improve athlete performance.

Existing systems mainly store performance data but provide limited intelligent analysis and personalized recommendations. Without AI-driven assistance, identifying strengths, weaknesses, injury risks, and performance trends becomes time-consuming and less effective.

The Objective

Build an AI-powered Athlete Performance Analysis Assistant that evaluates athlete performance using training records, fitness metrics, and match statistics.

The solution should:

- Retrieve athlete statistics, training records, and fitness metrics using Retrieval-Augmented Generation (RAG).
- Analyze historical and current performance to identify strengths, weaknesses, and improvement opportunities.
- Generate personalized training schedules and recovery recommendations.
- Produce AI-powered coaching reports and performance summaries.
- Implement a Multi-Agent System consisting of:
 - Athlete Data Agent
 - Performance Analysis Agent
 - Training Recommendation Agent
 - Coaching Insight Agent
- Display athlete performance trends, fitness metrics, comparisons, and recommendations through an interactive dashboard.